

MATH 1210 Worksheet 7 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

1. (a) Is an identity matrix invertible? If yes, what is its inverse?

Solution: The identity matrix I_n of size n is invertible, and its inverse is $I_n^{-1} = I_n$. This follows from the fact that

$$I_n \cdot I_n = I_n.$$

- (b) Is a square zero matrix invertible? If yes, what is its inverse?

Solution: Suppose that $A = 0_{n \times n}$ is the zero matrix of size n . Suppose that it has an inverse A^{-1} . This inverse should satisfy

$$AA^{-1} = A^{-1}A = I_n$$

by definition of the inverse. On the other hand, since A is the zero matrix we would also have

$$AA^{-1} = 0_{n \times n}A^{-1} = 0_{n \times n}.$$

Combining the two would give $I_n = 0_{n \times n}$, which is absurd. Hence a zero matrix is not invertible.

- (c) Suppose that A is a square matrix such that $A^n = 0$ for some integer $n \geq 2$. Show that A is not invertible.

Solution: By part b), the matrix A^n is not invertible. Hence its determinant is nonzero:

$$|A^n| = 0.$$

Since the determinant is multiplicative, we have $0 = |A^n| = |A|^n$. This implies that $|A| = 0$, hence A is not invertible.

2. For a real number t consider the matrix

$$A = \begin{pmatrix} t & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & -1 \end{pmatrix}.$$

- (a) Find the value(s) of t for which A is invertible.

Solution: By expanding along the first row we find that the determinant of A is

$$\begin{vmatrix} t & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & -1 \end{vmatrix} = t \begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} + 6 \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = -7t + 6.$$

The matrix A is invertible precisely when the determinant is nonzero, which is the case when $t \neq \frac{6}{7}$.

- (b) Use the adjoint method to compute A^{-1} for $t = 1$.

Solution: For $t = 1$ the matrix is

$$A = \begin{pmatrix} 1 & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & -1 \end{pmatrix}.$$

We compute the nine cofactors of A :

$$\begin{aligned} c_{11} &= \begin{vmatrix} 3 & 4 \\ 1 & -1 \end{vmatrix} = -7, & c_{12} &= -\begin{vmatrix} 1 & 4 \\ 0 & -1 \end{vmatrix} = 1, & c_{13} &= \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1 \\ c_{21} &= -\begin{vmatrix} 0 & 6 \\ 1 & -1 \end{vmatrix} = 6, & c_{22} &= \begin{vmatrix} 1 & 6 \\ 0 & -1 \end{vmatrix} = -1, & c_{23} &= -\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = -1 \\ c_{31} &= \begin{vmatrix} 0 & 6 \\ 3 & 4 \end{vmatrix} = -18, & c_{32} &= -\begin{vmatrix} 1 & 6 \\ 1 & 4 \end{vmatrix} = 2, & c_{33} &= \begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix} = 3. \end{aligned}$$

Hence the matrix of cofactors is

$$C = \begin{pmatrix} -7 & 1 & 1 \\ 6 & -1 & -1 \\ -18 & 2 & 3 \end{pmatrix}.$$

The adjoint of A is then

$$\text{adj}(A) = C^T = \begin{pmatrix} -7 & 6 & -18 \\ 1 & -1 & 2 \\ 1 & -1 & 3 \end{pmatrix}.$$

By the first part we know that $|A| = -1$, thus

$$A^{-1} = -\text{adj}(A) = \begin{pmatrix} 7 & -6 & 18 \\ -1 & 1 & -2 \\ -1 & 1 & -3 \end{pmatrix}.$$

- (c) Use the direct method to compute A^{-1} for $t = 1$.

Solution: Now for the direct method we use Gauss-Jordan elimination

$$\begin{aligned}
& \left(\begin{array}{ccc|ccc} 1 & 0 & 6 & 1 & 0 & 0 \\ 1 & 3 & 4 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right) R_2 \rightarrow R_2 - R_1 \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 6 & 1 & 0 & 0 \\ 0 & 3 & -2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right) R_2 \rightarrow R_2 - 3R_3 \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 6 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & -3 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right) R_3 \leftrightarrow R_2 \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 6 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 & -3 \end{array} \right) R_2 \rightarrow R_3 + R_2 \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & -2 \\ 0 & 0 & 1 & -1 & 1 & -3 \end{array} \right) R_1 \rightarrow R_1 - 6R_3 \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -6 & 18 \\ 0 & 1 & 0 & -1 & 1 & -2 \\ 0 & 0 & 1 & -1 & 1 & -3 \end{array} \right) R_1 \rightarrow R_1 - 6R_3.
\end{aligned}$$

We find

$$A^{-1} = \begin{pmatrix} 7 & -6 & 18 \\ -1 & 1 & -2 \\ -1 & 1 & -3 \end{pmatrix}.$$

3. Consider the matrix

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 5 & 0 & 3 \\ t & 2 & 2 \end{pmatrix}$$

where t is a real number. It is given to you that A is invertible and that the entry in the second column and third row of A^{-1} is $1/7$. Use the adjoint method to find t .

Solution: By the adjoint method

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{|A|} C^T = \frac{1}{|A|} \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix}.$$

where

$$C = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

is the matrix of cofactors of A . It is given to us that

$$\frac{c_{23}}{|A|} = \frac{1}{7}.$$

This is equivalent to the equation

$$7c_{23} = |A|. \tag{1}$$

The cofactor c_{23} of A is given by

$$c_{23} = -m_{23} = -\begin{vmatrix} 2 & 1 \\ t & 2 \end{vmatrix} = -(4 - t) = t - 4.$$

By expanding along the second row we find that the determinant of A is

$$\begin{vmatrix} 2 & 1 & -1 \\ 5 & 0 & 3 \\ t & 2 & 2 \end{vmatrix} = -5 \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ t & 2 \end{vmatrix} = -32 + 3t.$$

If we substitute the cofactor and the determinant in equation (1) we obtain the following equation that we solve for t :

$$\begin{aligned} 7(t - 4) &= -32 + 3t \\ \iff 7t - 28 &= -32 + 3t \\ \iff 4t &= -4 \\ \iff t &= -1. \end{aligned}$$

4. (a) Use any method to compute the inverse of

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 2 & 1 \\ 5 & 1 & -1 \end{pmatrix}.$$

Solution: We use the adjoint method to find A^{-1} . First we compute the nine cofactors of A :

$$\begin{aligned} c_{11} &= \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3, & c_{12} &= -\begin{vmatrix} 0 & 1 \\ 5 & -1 \end{vmatrix} = 5, & c_{13} &= \begin{vmatrix} 0 & 2 \\ 5 & 1 \end{vmatrix} = -10 \\ c_{21} &= -\begin{vmatrix} 3 & 1 \\ 1 & -1 \end{vmatrix} = 4, & c_{22} &= \begin{vmatrix} 2 & 1 \\ 5 & -1 \end{vmatrix} = -7, & c_{23} &= -\begin{vmatrix} 2 & 3 \\ 5 & 1 \end{vmatrix} = 13 \\ c_{31} &= \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} = 1, & c_{32} &= -\begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = -2, & c_{33} &= \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = 4. \end{aligned}$$

Hence the matrix of cofactors is

$$C = \begin{pmatrix} -3 & 5 & -10 \\ 4 & -7 & 13 \\ 1 & -2 & 4 \end{pmatrix}.$$

The adjoint of A is then

$$\text{adj}(A) = C^T = \begin{pmatrix} -3 & 4 & 1 \\ 5 & -7 & -2 \\ -10 & 13 & 4 \end{pmatrix}.$$

Then we compute the determinant by expanding along the first column

$$|A| = 2 \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} + 5 \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} = 2(-2 - 1) + 5(3 - 2) = -1.$$

Thus the inverse is given by

$$A^{-1} = -\text{adj}(A) = \begin{pmatrix} 3 & -4 & -1 \\ -5 & 7 & 2 \\ 10 & -13 & -4 \end{pmatrix}.$$

(b) Use the result of part (a) to solve the system

$$\begin{aligned} 2x + 5z &= 2 \\ 3x + 2y - z &= -1 \\ 5x + y - z &= -1 \end{aligned}$$

Solution: In matrix form the system can be written as

$$\begin{pmatrix} 2 & 0 & 5 \\ 3 & 2 & -1 \\ 5 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}.$$

We recognize the coefficient matrix of the system is A^T . Hence the system is $A^T X = B$ where

$$B = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

and has a unique solution given by

$$X = (A^T)^{-1} B = (A^{-1})^T B.$$

Using part (a) we find that

$$(A^{-1})^T = \begin{pmatrix} 3 & -5 & 10 \\ -4 & 7 & -13 \\ -1 & 2 & -4 \end{pmatrix}.$$

Thus we get

$$X = \begin{pmatrix} 3 & -5 & 10 \\ -4 & 7 & -13 \\ -1 & 2 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}.$$

(c) Are the vectors $\mathbf{u}_1 = \langle 2, 0, 5 \rangle$, $\mathbf{u}_2 = \langle 3, 2, 1 \rangle$ and $\mathbf{u}_3 = \langle 1, 1, -1 \rangle$ linearly dependent?

Solution: No. The columns of A are precisely the vectors $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$. Since the matrix is invertible, the determinant is nonzero and the vectors are linearly independent.

5. Consider the vectors $\mathbf{u}_1 = \langle 1, 2, 3, 6 \rangle$, $\mathbf{u}_2 = \langle 5, -2, 5, -4 \rangle$, $\mathbf{u}_3 = \langle 3, 3, 2, 1 \rangle$ and

$$\mathbf{u}_4 = 2\mathbf{u}_1 - 3\mathbf{u}_2 + \mathbf{u}_3 = \langle -10, 13, -7, 25 \rangle.$$

Without doing any computation, is the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 & 6 \\ 5 & -2 & 5 & -4 \\ 3 & 3 & 2 & 1 \\ -10 & 13 & -7 & 25 \end{pmatrix}$$

invertible?

Solution: The rows of A are the vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4$. Hence the columns of A^T are $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4$. Since the vectors are linearly dependent (we can write $\mathbf{u}_4$ as a linear combination of the other vectors), the determinant of A^T is 0. Hence A^T is not invertible. This implies that A is not invertible, since otherwise we would have $(A^T)^{-1} = (A^{-1})^T$.

6. Consider the linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ sending $\langle 1, 1 \rangle$ to $\langle -1, 4 \rangle$, and $\langle 0, 1 \rangle$ to $\langle -2, 3 \rangle$.

(a) Use linearity to find the image of $\langle 1, 0 \rangle$.

Solution: We know that $T(\langle 1, 1 \rangle) = \langle -1, 4 \rangle$ and $T(\langle 0, 1 \rangle) = \langle -2, 3 \rangle$. Moreover, we have

$$\langle 1, 0 \rangle = \langle 1, 1 \rangle - \langle 0, 1 \rangle.$$

hence

$$T(\langle 1, 0 \rangle) = T(\langle 1, 1 \rangle) - T(\langle 0, 1 \rangle) = \langle -1, 4 \rangle - \langle -2, 3 \rangle = \langle 1, 1 \rangle.$$

(b) Find the associated matrix.

Solution: The columns of the associated matrix are exactly $T(\langle 1, 0 \rangle)$ and $T(\langle 0, 1 \rangle)$. Hence, by part (a) the associated matrix is

$$\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix}.$$

(c) Find the eigenvalues and eigenvectors of T .

Solution: The eigenvalues of T are the roots of the characteristic equation:

$$|A - \lambda I_2| = 0,$$

where A is the associated matrix. We compute the determinant

$$|A - \lambda I_2| = \begin{vmatrix} 1 - \lambda & -2 \\ 1 & 3 - \lambda \end{vmatrix} = \lambda^2 - 4\lambda + 5.$$

Hence the eigenvalues are the roots of $\lambda^2 - 4\lambda + 5 = 0$:

$$\lambda = \frac{4 \pm \sqrt{(-4)^2 - 4 \cdot 5}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i.$$

The eigenvectors $\langle v_1, v_2 \rangle$ to the eigenvalue $\lambda = 2 - i$ are the solutions of the homogeneous system $(A - (2 - i)I_2)\mathbf{v} = 0$, which can be written as

$$\begin{pmatrix} -(1 - i) & -2 \\ 1 & 1 + i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

We bring the augmented matrix in RREF by Gauss-Jordan elimination:

$$\begin{aligned} \left(\begin{array}{cc|c} -(1-i) & -2 & 0 \\ 1 & 1+i & 0 \end{array} \right) R_1 \rightarrow R_2 &\rightarrow \left(\begin{array}{cc|c} 1 & 1+i & 0 \\ -(1+i) & -2 & 0 \end{array} \right) R_2 \rightarrow R_2 + (1-i)R_1 \\ &\rightarrow \left(\begin{array}{cc|c} 1 & 1+i & 0 \\ 0 & 0 & 0 \end{array} \right) R_2 \rightarrow R_2 - (1-i)R_1 \end{aligned}$$

Hence $v_1 + (1 + i)v_2 = 0$, equivalently $v_1 = -(1 + i)v_2$. Setting the parameter $t = v_2$, the eigenvectors to the eigenvalue $\lambda = (2 - i)$ are

$$\mathbf{v} = \begin{pmatrix} -t(1 + i) \\ t \end{pmatrix} = t \begin{pmatrix} -1 - i \\ 1 \end{pmatrix}$$

with t nonzero. A similar computation shows that the eigenvectors to the eigenvalue $\lambda = 2 + i$ are

$$\mathbf{v} = t \begin{pmatrix} -1 + i \\ 1 \end{pmatrix}$$

with t nonzero. Alternatively, one can find these eigenvectors by using the the following seen in class: if A is a real matrix and $\mathbf{v}$ is an eigenvector to the eigenvalue λ , then the complex conjugate $\bar{\mathbf{v}}$ is an eigenvector to the complex conjugate eigenvalue $\bar{\lambda}$.

7. Consider the linear map $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ given by

$$\begin{aligned} v'_1 &= v_1 + 2v_2 - v_3 \\ v'_2 &= 5v_1 + 4v_2 + v_3 \\ v'_3 &= 2v_1 + v_2. \end{aligned}$$

(a) Compute $T(\mathbf{v})$ for $\mathbf{v} = \langle 1, 2, -1 \rangle$.

Solution: We substitute $\mathbf{v}$ in the definition of T and find

$$\begin{aligned} v'_1 &= 1 + 2 \cdot 2 - (-1) = 6 \\ v'_2 &= 5 + 4 \cdot 2 + (-1) = 8 \\ v'_3 &= 2 + 2 = 4. \end{aligned}$$

(b) Find $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ such that $T(\mathbf{v}) = \langle 3, 3, -1 \rangle$.

Solution: Finding $\mathbf{v}$ such that $T(\mathbf{v}) = \langle 3, 3, -1 \rangle$ means solving the system

$$\begin{aligned} 3 &= v_1 + 2v_2 - v_3 \\ 3 &= 5v_1 + 4v_2 + v_3 \\ -1 &= 2v_1 + v_2. \end{aligned}$$

We bring the augmented matrix in RREF using Gauss-Jordan elimination:

$$\begin{aligned}
 \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 5 & 4 & 1 & 3 \\ 2 & 1 & 0 & -1 \end{array} \right) & \begin{array}{l} R_2 \rightarrow R_2 - 5R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 6 & -12 \\ 0 & -3 & 2 & -7 \end{array} \right) \begin{array}{l} R_2 \rightarrow -R_2/6 \\ \\ \end{array} \\
 & \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & -3 & 2 & -7 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - 2R_2 \\ R_3 \rightarrow 3R_2 + R_3 \end{array} \\
 & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -1 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - R_3 \\ \\ \end{array} \\
 & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -1 \end{array} \right) \begin{array}{l} \\ \\ R_3 \rightarrow -R_3 \end{array} \\
 & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right).
 \end{aligned}$$

Hence $\mathbf{v} = \langle -2, 3, 1 \rangle$.

8. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation such that $T(2\mathbf{i}) = \langle 2, -4, 6 \rangle$, $T(-\mathbf{j}) = \langle 3, 1, -2 \rangle$ and $T(\mathbf{v}) = \langle 1, 0, 5 \rangle$ for $\mathbf{v} = \langle 1, -1, 1 \rangle$.

- (a) Use the linearity of T to find $T(\mathbf{k})$.

Solution: By linearity we have $T(\mathbf{i}) = \langle 1, -2, 3 \rangle$ since

$$2T(\mathbf{i}) = T(2\mathbf{i}) = \langle 2, -4, 6 \rangle.$$

Similarly we get $T(\mathbf{j}) = \langle -3, -1, 2 \rangle$. In terms of the basis vectors we can write $\mathbf{v}$ as $\mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, so that

$$\mathbf{v} - \mathbf{i} + \mathbf{j} = \mathbf{k}.$$

By the linearity of T we get that

$$\begin{aligned}
 T(\mathbf{k}) &= T(\mathbf{v} - \mathbf{i} + \mathbf{j}) \\
 &= T(\mathbf{v}) - T(\mathbf{i}) + T(\mathbf{j}) \\
 &= \langle 1, 0, 5 \rangle - \langle 1, -2, 3 \rangle + \langle -3, -1, 2 \rangle \\
 &= \langle -3, 1, 4 \rangle
 \end{aligned}$$

- (b) Find the matrix A associated with T .

Solution: The columns of the matrix associated with T are $T(\mathbf{i}), T(\mathbf{j})$ and $T(\mathbf{k})$. Hence

$$A = \begin{pmatrix} 1 & -3 & -3 \\ -2 & -1 & 1 \\ 3 & 2 & 4 \end{pmatrix}.$$

9. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that reflects every vector in the plane $x = y$.

- (a) Find the associated matrix.

Solution: To find the associated matrix, we need to determine the image of the vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$. The plane $x = y$ is the plane containing the z -axis and intersecting the xy -plane along the line $x = t, y = t$ (see figure (??) on the last page). Since the vector $\mathbf{k}$ is parallel to the plane, the reflection of $\mathbf{k}$ along the plane is $\mathbf{k}$. The reflection of $\mathbf{i}$ along this plane is $\mathbf{j}$ and the reflection of $\mathbf{j}$ is $\mathbf{i}$. So we have $T(\mathbf{i}) = \mathbf{j}$, $T(\mathbf{j}) = \mathbf{i}$ and $T(\mathbf{k}) = \mathbf{k}$. It follows that the associated matrix is

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (b) Find the image of the vector $\mathbf{v} = \langle 1, 2, 3 \rangle$ under T .

Solution: The image is

$$T(\mathbf{v}) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

- (c) Find $\mathbf{v}$ such that $T(\mathbf{v}) = \langle 2, 0, 1 \rangle$.

Solution: The linear transformation permutes the x and y components. Hence, for $\mathbf{v} = \langle 0, 2, 1 \rangle$ we have $T(\mathbf{v}) = \langle 2, 0, 1 \rangle$. Alternatively, one can simply solve the corresponding system of linear equations.

10. Consider the map that sends a nonzero vector $\mathbf{v}$ to the unit vector in the opposite direction to $\mathbf{v}$. Determine if the map is linear.

Solution: The map is not linear. First we note that the map is given by $T(\mathbf{v}) = -\frac{\mathbf{v}}{|\mathbf{v}|}$. Now take the vectors $\mathbf{u}_1 = \langle 1, 1, 0 \rangle$ and $\mathbf{u}_2 = \langle -1, 1, 0 \rangle$. On the one hand we have

$$T(\mathbf{u}_1) = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\rangle$$

$$T(\mathbf{u}_2) = \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\rangle.$$

Hence

$$T(\mathbf{u}_1) + T(\mathbf{u}_2) = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\rangle + \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right\rangle = \langle 0, -\sqrt{2}, 0 \rangle.$$

On the other hand we have

$$T(\mathbf{u}_1 + \mathbf{u}_2) = T(\langle 0, 2, 0 \rangle) = \langle 0, -1, 0 \rangle.$$

Since

$$T(\mathbf{u}_1 + \mathbf{u}_2) \neq T(\mathbf{u}_1) + T(\mathbf{u}_2),$$

the map is not linear.

11. Consider the matrix

$$A = \begin{pmatrix} -1 & 0 & t \\ 3 & 3 & 3 \\ 2 & 0 & 5 \end{pmatrix}$$

where t is a real number.

- (a) Without doing any computation, explain why $\lambda = 3$ is always eigenvalue, for any value of t . Find all the corresponding eigenvectors.

Solution: We can observe that the vector $\mathbf{v} = \langle 0, 1, 0 \rangle$ satisfies

$$\begin{pmatrix} -1 & 0 & t \\ 3 & 3 & 3 \\ 2 & 0 & 5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = 3 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Hence it is an eigenvector to the eigenvalue $\lambda = 3$, and the other eigenvectors are of the form

$$\mathbf{v} = \begin{pmatrix} 0 \\ t \\ 0 \end{pmatrix}$$

with t nonzero.

- (b) Find the other eigenvalues when $t = 2$.

Solution: The eigenvalues λ are the roots of the characteristic equation

$$|A - \lambda I_3| = 0.$$

We compute (expanding along the second column):

$$\begin{aligned} |A - \lambda I_3| &= \begin{vmatrix} -1 - \lambda & 0 & 2 \\ 3 & 3 - \lambda & 3 \\ 2 & 0 & 5 - \lambda \end{vmatrix} \\ &= (3 - \lambda) \begin{vmatrix} -1 - \lambda & 2 \\ 2 & 5 - \lambda \end{vmatrix} \\ &= (3 - \lambda) (-(1 + \lambda)(5 - \lambda) - 4) \\ &= (3 - \lambda)(\lambda^2 - 4\lambda - 9). \end{aligned}$$

Hence the eigenvalues are $\lambda = 3$ (found in (a)) and the roots of the quadratic equation $\lambda^2 - 4\lambda - 9 = 0$:

$$\lambda = \frac{4 \pm \sqrt{(-4)^2 - 4 \cdot (-9)}}{2} = \frac{4 \pm \sqrt{52}}{2} = 2 \pm \sqrt{13}.$$

- (c) For which value(s) of t is $\langle 1, 3, 2 \rangle$ an eigenvector to the eigenvalue $\lambda = 6$?

Solution: If $\langle 1, 3, 2 \rangle$ is an eigenvector to the eigenvalue $\lambda = 1$, then

$$\begin{pmatrix} -1 & 0 & t \\ 3 & 3 & 3 \\ 2 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = 6 \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 18 \\ 12 \end{pmatrix}. \quad (2)$$

The left hand side of this equation is

$$\begin{pmatrix} -1 & 0 & t \\ 3 & 3 & 3 \\ 2 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 + 2t \\ 18 \\ 12 \end{pmatrix}.$$

Comparing the LHS and RHS of (2) we find that $-1 + 2t = 6$, so $t = 7/2$.

12. Find all eigenvalues of the matrix $A = \begin{pmatrix} 3 & 5 & -5 \\ -2 & 3 & 2 \\ -2 & 5 & 0 \end{pmatrix}$. Find the eigenvectors to the largest eigenvalue.

Solution: The eigenvalues λ are the roots of the characteristic equation

$$|A - \lambda I_3| = 0.$$

We compute (expanding along the first row):

$$\begin{aligned} |A - \lambda I_3| &= \begin{vmatrix} 3 - \lambda & 5 & -5 \\ -2 & 3 - \lambda & 2 \\ -2 & 5 & -\lambda \end{vmatrix} \\ &= (3 - \lambda) \begin{vmatrix} 3 - \lambda & 2 \\ 5 & -\lambda \end{vmatrix} - 5 \begin{vmatrix} -2 & 2 \\ -2 & -\lambda \end{vmatrix} - 5 \begin{vmatrix} -2 & 3 - \lambda \\ -2 & 5 \end{vmatrix} \\ &= (3 - \lambda) (\lambda^2 - 3\lambda - 10) - 5(2\lambda + 4) - 5(-10 + 6 - 2\lambda) \\ &= (3 - \lambda) (\lambda^2 - 3\lambda - 10). \end{aligned}$$

Hence the eigenvalues are $\lambda = 3$ and the roots of the quadratic equation $\lambda^2 - 3\lambda - 10 = 0$:

$$\lambda = \frac{3 \pm \sqrt{(-3)^2 - 4 \cdot (-10)}}{2} = \frac{3 \pm \sqrt{49}}{2} = \frac{3 \pm 7}{2}.$$

In conclusion the three eigenvalues are

$$\{-2, 3, 5\}.$$

The eigenvectors $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ corresponding to the largest eigenvalue $\lambda = 5$ are the solutions of the homogeneous system $(A - 5I_3)\mathbf{v} = 0$, which can be written as

$$\begin{pmatrix} -2 & 5 & -5 \\ -2 & -2 & 2 \\ -2 & 5 & -5 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We bring the augmented matrix in RREF by Gauss-Jordan elimination:

$$\begin{aligned} \left(\begin{array}{ccc|c} -2 & 5 & -5 & 0 \\ -2 & -2 & 2 & 0 \\ -2 & 5 & -5 & 0 \end{array} \right) & \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|c} -2 & 5 & -5 & 0 \\ 0 & -7 & 7 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} R_2 \rightarrow -R_2/7 \\ \\ \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} -2 & 5 & -5 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow R_1 - 5R_2 \\ \\ \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} R_1 \rightarrow -R_1/2 \\ \\ \end{array} \\ & \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) RREF \end{aligned}$$

Hence $v_1 = 0$ and $v_2 - v_3 = 0$, equivalently $v_2 = v_3$. Setting the parameter $t = v_3$, the eigenvectors to the eigenvalue $\lambda = 5$ are

$$\mathbf{v} = \begin{pmatrix} 0 \\ t \\ t \end{pmatrix}$$

with t nonzero.